Multi-layer perceptrons

Plan

1. **Representation and structure**
2. Learning: Backpropagation
3. Optimisation of learning

Neural networks I

- Logistic regression is the simplest neural network:
  - single layer
  - *model*
    - binary-class case: $P'(C_1|\mathbf{x}) = \sigma(\mathbf{w}^T\mathbf{x})$, single hidden unit, learn $\mathbf{w}$.
    - K classes case: $P'(C_i|\mathbf{x}) = \frac{\exp \mathbf{w}_i^T\mathbf{x}}{\sum_{j=1}^{K} \exp \mathbf{w}_j^T\mathbf{x}}$, K hidden units, learn $\{\mathbf{w}_i | i = 1...k\}$.
  - *error function*, Cross-entropy:

$$L(\mathbf{y}, \mathbf{x}) = -\sum_{j=1}^{K} P(C_j|\mathbf{x}) \ln P'(C_j|\mathbf{x}) = -\sum_{j=1}^{K} y_j \ln P'(C_j|\mathbf{x})$$

$\mathbf{y}$ is the one-hot vector encoding of the class label.

Neural networks II

- a less trivial NN is a repetitive application of logistic regression with the error function at the last layer.
- at the heart of NNs we have function composition
- To do optimization over function composition we make heavy use of the *chain rule*
- The chain rule comes under the name *backpropagation* in neural networks.

Chain rule, reminder, I

- let:

$$\begin{aligned} f(\mathbf{x}) &: \mathbf{x} \rightarrow y, \mathbf{x} \in R^d, y \in R \\ g(y) &: y \rightarrow z, y \in R, z \in R \end{aligned}$$

- The partial derivative of the composition $z = g(f(\mathbf{x}))$ with the ith dimension of $\mathbf{x}$ is given by:

$$\frac{\partial z}{\partial x_i}$$

Chain rule, reminder, I

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$$

- The partial derivative of the composition $z = g(f(\mathbf{x}))$ with the ith dimension of $\mathbf{x}$ is given by:

$$
\begin{aligned}
\frac{\partial z}{\partial x_i} &= \frac{\partial z}{\partial y}\frac{\partial y}{\partial x_i} \\
&= \frac{\partial g(y)}{\partial y}\frac{\partial f(\mathbf{x})}{\partial x_i}
\end{aligned}
$$

Chain rule, reminder, II

- what happens when y is a vector? i.e. let

$$\begin{aligned} f(\mathbf{x}) &: \mathbf{x} \rightarrow \mathbf{y}, \mathbf{x} \in R^d, \mathbf{y} \in R^n \\ g(\mathbf{y}) &: \mathbf{y} \rightarrow z, \mathbf{y} \in R^n, z \in R \end{aligned}$$

- What is the meaning of $\frac{\partial z}{\partial x_i}$?
  - describes the sensitivity of z to x_i
- In how many ways does x_i affect z?
  - in n ways. Which ones?

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  - all $y_j, i = j...n$
- Thus to establish the sensitivity/partial derivative of z with x_i we need all of them:

$$\frac{\partial z}{\partial x_i} =$$

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- Thus to establish the sensitivity/partial derivative of z with x_i we need all of them:

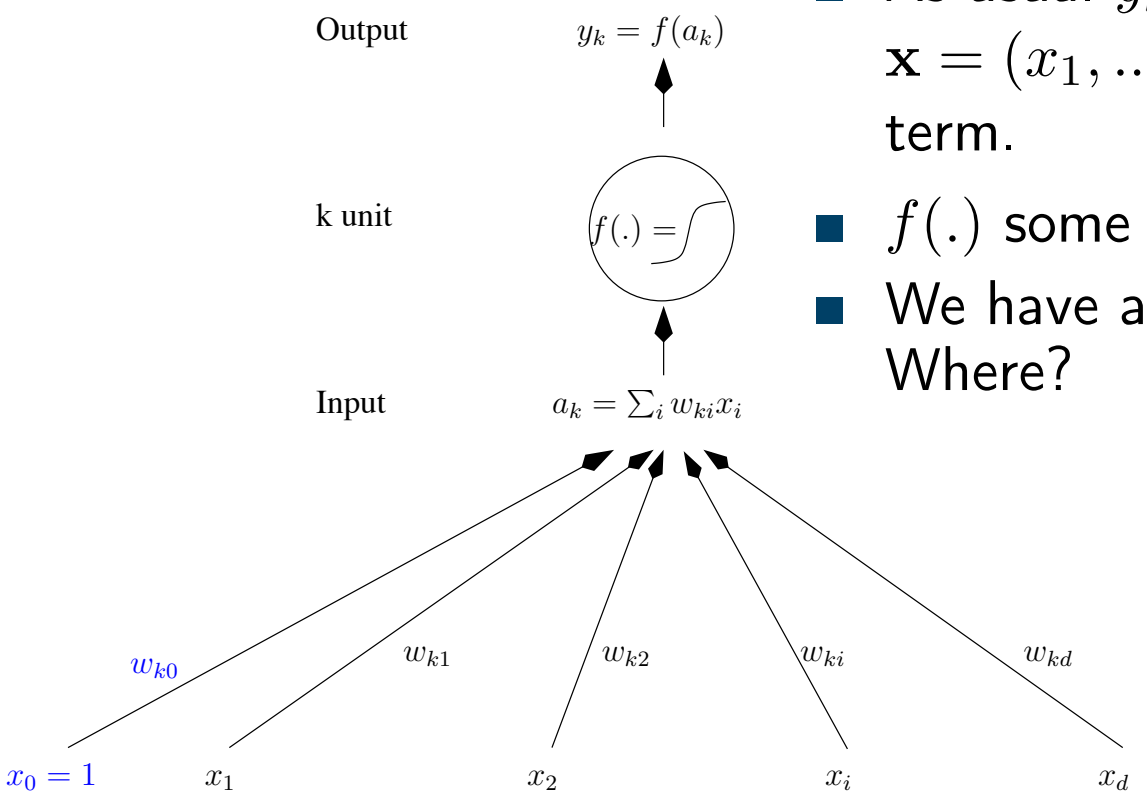
$$\frac{\partial z}{\partial x_i} = \frac{\partial z}{\partial y_1}\frac{\partial y_1}{\partial x_i} + \frac{\partial z}{\partial y_2}\frac{\partial y_2}{\partial x_i} + ... + \frac{\partial z}{\partial y_n}\frac{\partial y_n}{\partial x_i}$$

the *multivariate chain rule*

NNs very short summary

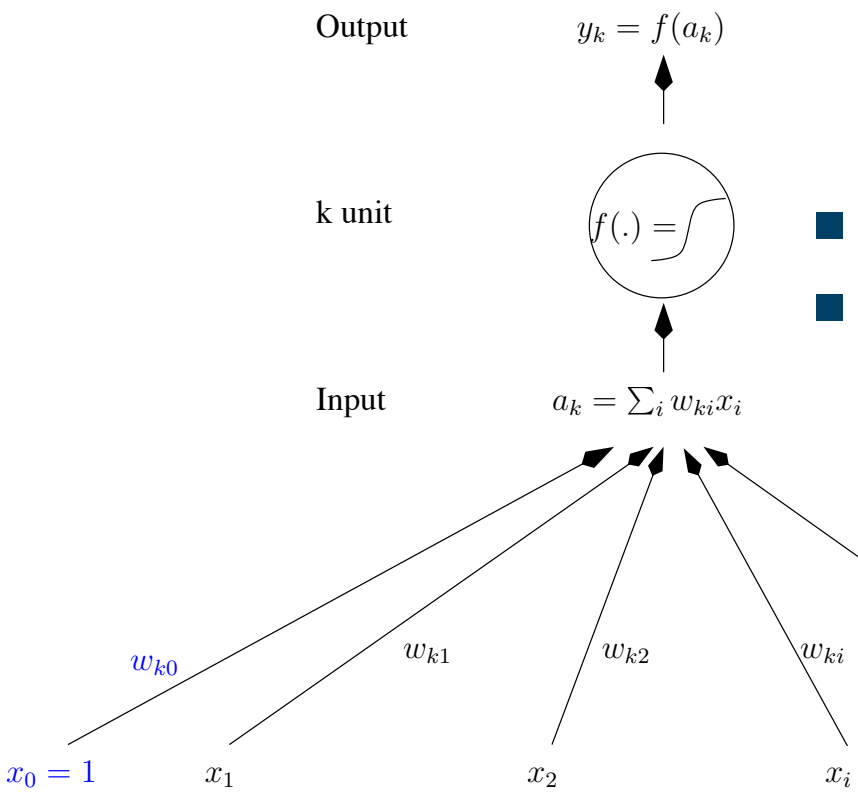
- standard NNs can be thought as an hierarchical composition of logistic regression models, loss function at the last layer.
- To compute the gradient of the loss
  - we rely on the chain rule to address the composition and the fact that indermediate results will be vectors
- NN course could stop here.

Single Layer Networks



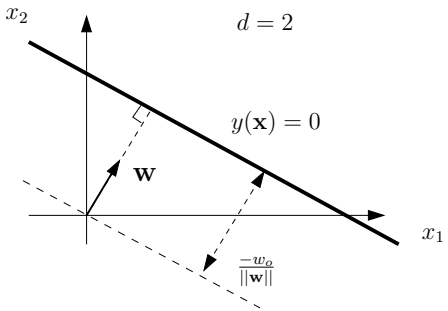
- As usual y_k the kth output, $\mathbf{x} = (x_1, ..., x_d)^T$ the input, w_{k0} the bias term.
- $f(.)$ some transformation function
- We have already seen such structures. Where?

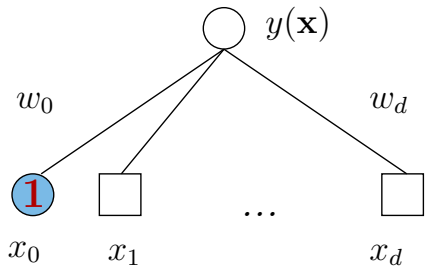
Single Layer Networks



- As usual y_k the kth output, $\mathbf{x} = (x_1, ..., x_d)^T$ the input, w_{k0} the bias term.
- $f(.)$ some transformation function
- We have already seen such structures. Where?
  - Perceptron: $f(a) = sign(a), k = 1$
  - Logistic regression :
    - $f(a) = \frac{1}{1+e^{-a}}, k = 1$
    - $f(a_k) = \frac{\exp a_k}{\sum_k \exp a_k}, k \in [1...c]$
  - Fisher's linear discriminant $f(.) = I(.)$

Limitations of perceptron



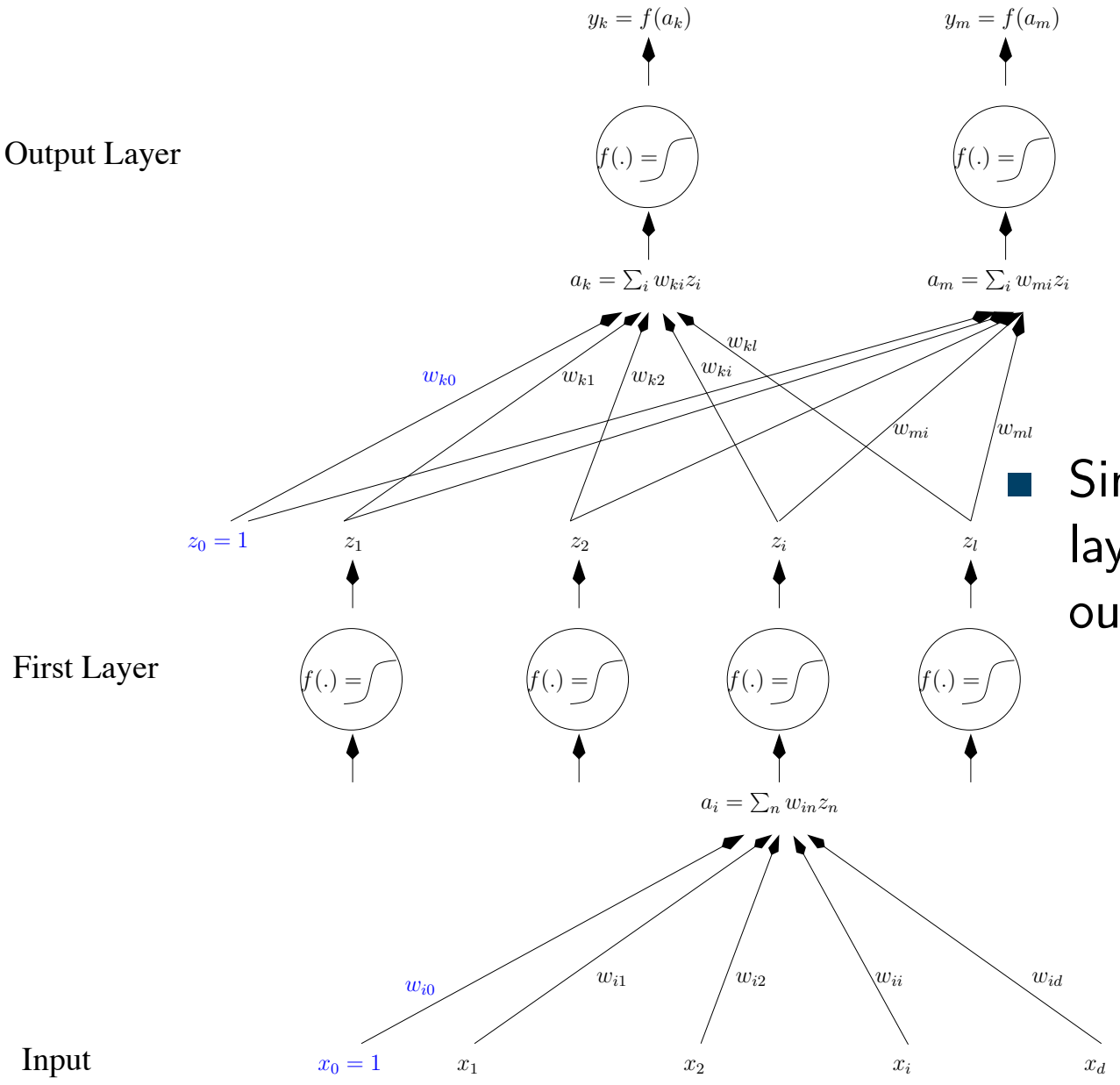


- It can only work on linearly seperable problems
- Most problems are not linearly seperable
- To get more complex surfaces we can supperimpose many nodes that use *non-linear* activation functions
- One layer: a set of connections/weights followed by the respective nodes.

MLP structure I

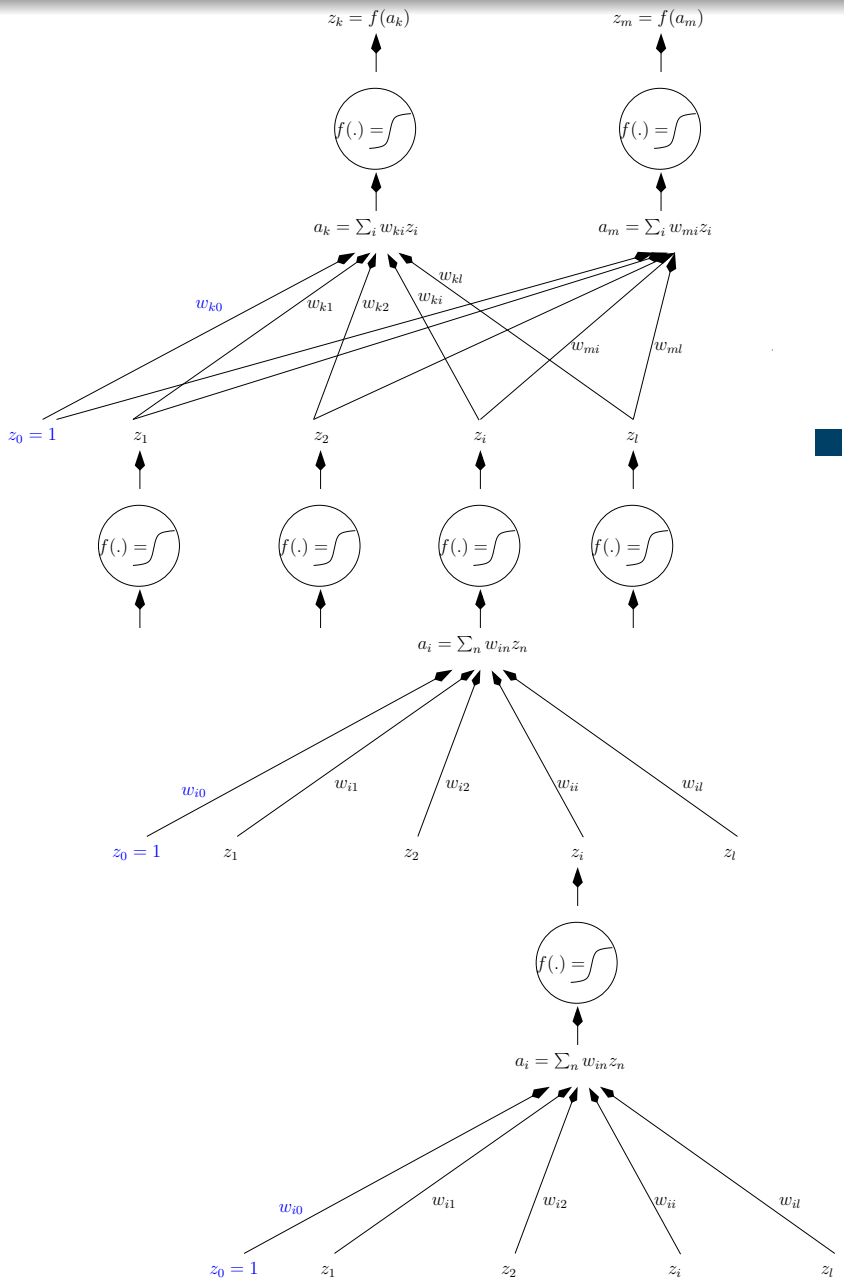
- Multi-Layer Perceptrons (MLP) = networks of many layers
- acyclic (feedforward): no backward connections
- completely connected: each node of a layer is connected to all nodes of the previous layer

MLP structure II



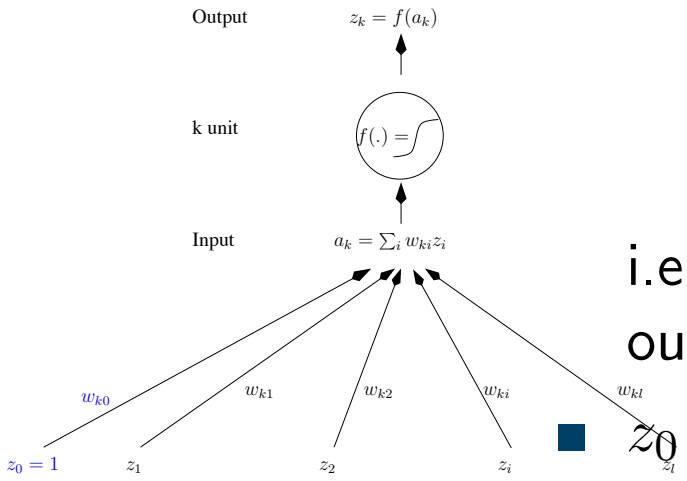
- Simplest example : 2 layers one hidden, one output

MLP structure III



- Nowdays five-layer, and more, nets are common: *deep neural networks* and *deep learning*. For example:
  - [2] use eight layers with 60 million parameters.
  - deep face, [3], 120M parameters, and [1], 10B parameters.

The hidden unit



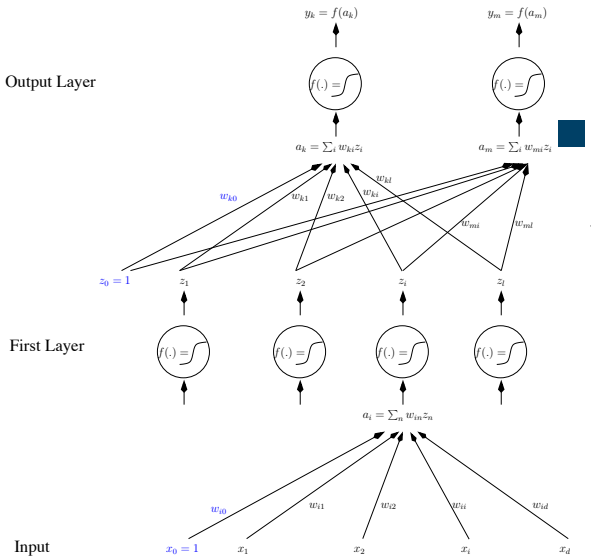
- Each unit k in some layer receives an activation value a_k given by:

$$a_k = \sum_i^l w_{ki} z_i = \mathbf{w}_k^T \mathbf{z}$$

i.e. a_k is a weighted combination of the outputs of the l units of the previous layer.

- $z_0 = 1$ and w_{k0} are there to incorporate the bias in the model.
- The output z_k of the unit is a non-linear function $f(.)$ of the activation value

The output layer



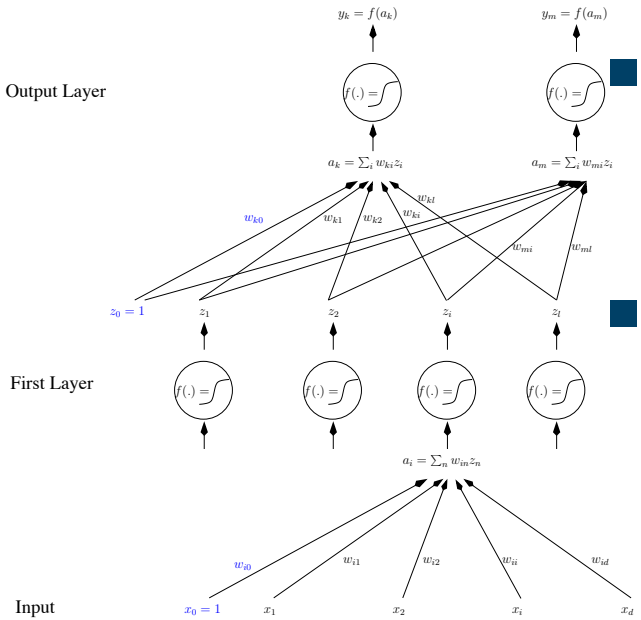
- Number of output units:
  - Regression: single unit corresponding to the target value
  - Binary classification: single unit corresponding to the "positive" class
  - Multi-class classification: one unit/class
- Similar to the hidden layers each k output unit gets as activation the linear combination of the outputs of the units of the previous layer:

$$a_k = \sum_i^l w_{ki} z_i$$

- and produces as output the non-linear transformation:

$$y_k = f(a_k)$$

Analytical function of a two layer MLP



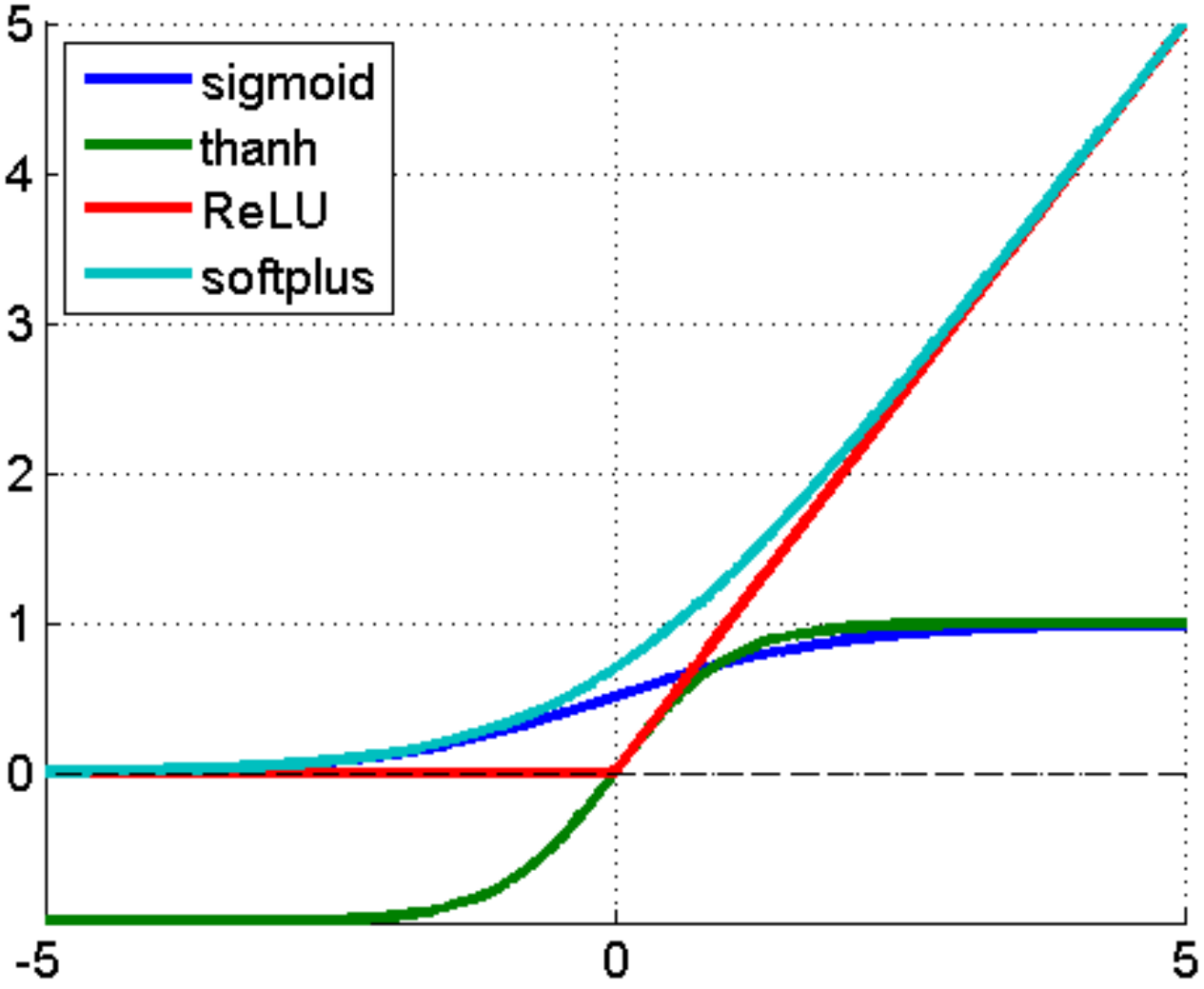
- An acyclic network can be seen as a composition of functional transformations of its inputs
- For example a two-layer MLP is:

$$y_k = g\left(\sum_{i=0}^{l} w_{ki}^{(2)} f\left(\sum_{n=0}^{d} w_{in}^{(1)} x_n\right)\right)$$

Activation functions I

- We need a *differentiable* function so that we can do gradient descent
- the activation function is typically *non-linear*
  - sigmoid $\sigma : \mathbb{R} \to [0,1]$: $f(a) = \frac{1}{1+e^{-a}}$
  - tanh: $\mathbb{R} \to [-1,1]$: $f(a) = \frac{e^a - e^{-a}}{e^a + e^{-a}}$
  - Rectified linear units (ReLU): $f(a) = \max(0, a)$
    - currently the most popular non-linearity,
    - introduces sparsity, no exploding or vanishing derivatives problem
  - soft plus : $f(a) = \ln(1 + e^x)$ smooth approximation of ReLU
- At the output layer the activation function can be:
  - regression : linear (identity)
  - classification: sigmoid or softmax

Activation functions II



Plan

1. Representation and structure
2. **Learning: Backpropagation**
3. Learning Optimisation

Error optimization in MLPs

- As always we are given some error function $E(\mathbf{w})$
- which we would like to optimise (minimise) with respect to the parameters of the model, here the vector of weights $\mathbf{w}$:

$$\mathbf{w} \equiv [w_{ij} | \text{weight of the connection of the } j \text{ unit to the } i \text{ unit}]$$

- The most classical way to do so for MLPs is by gradient descent

$$\begin{aligned} \nabla E[\mathbf{w}] &\equiv [\frac{\partial E}{\partial w_{10}}, \frac{\partial E}{\partial w_{11}}, \ldots, \frac{\partial E}{\partial w_{ij}}, \ldots \frac{\partial E}{\partial w_{ml}}] \\ \mathbf{w}^{(\tau+\mathbf{1})} &\leftarrow \mathbf{w}^{(\tau)} - \eta \nabla E[\mathbf{w}^{(\tau)}] \end{aligned}$$

η the learning rate

How to compute the gradient I

- We will make heavy use of the chain rule.
- The contribution of a weight w_{ij} to the error E comes only through:
  - the activation level $a_i = \sum_k w_{ik} z_k$
  - of the ith unit to which it connects the output z_j of the jth unit; Thus:

$$\frac{\partial E}{\partial w_{ij}} = \frac{\partial E}{\partial a_i} \frac{\partial a_i}{\partial w_{ij}} \tag{1}$$

- We have $a_i = \sum_k w_{ik} z_k \Rightarrow \frac{\partial a_i}{\partial w_{ij}} = z_j$
- And we set $\delta_i \equiv \frac{\partial E}{\partial a_i}$ (δ_i is called the error at the i node)
- Thus w_{ij} gradient can be written as:

$$\frac{\partial E}{\partial w_{ij}} = \delta_i z_j \tag{2}$$

How to compute the gradient II

- So the gradient of the error with respect to the w_{ij} weight that connects the ouput of unit j to the input of unit i is given according to equation:

$$\frac{\partial E}{\partial w_{ij}} = \delta_i z_j$$

  by the product of the output z_j of the unit j with the error δ_i of the i input unit.
- *Thus* to get the gradient we need to compute δ_i at each *hidden* and *output* unit.

Compute δ_k at the kth output unit

- We have:

$$
\begin{aligned}
\delta_k &= \frac{\partial E}{\partial a_k} \\
y_k &= f(a_k) \\
&\Rightarrow \\
\delta_k &= \frac{\partial E}{\partial y_k}\frac{\partial y_k}{\partial a_k} = \frac{\partial E}{\partial y_k}\frac{\partial f(a_k)}{\partial a_k} \\
&= \frac{\partial E}{\partial y_k} f'(a_k) \qquad (3)
\end{aligned}
$$

- thus we just need
  - the derivative of the error with the y_k output
  - and the derivative of the activation function evaluated at the input a_k of the k output unit.

Compute δ_j at the jth hidden unit

- We have: $\delta_j = \frac{\partial E}{\partial a_j}$
- a_j affects the error through the activation levels a_k of every unit k to which the output of the j unit is passed as input. Thus:

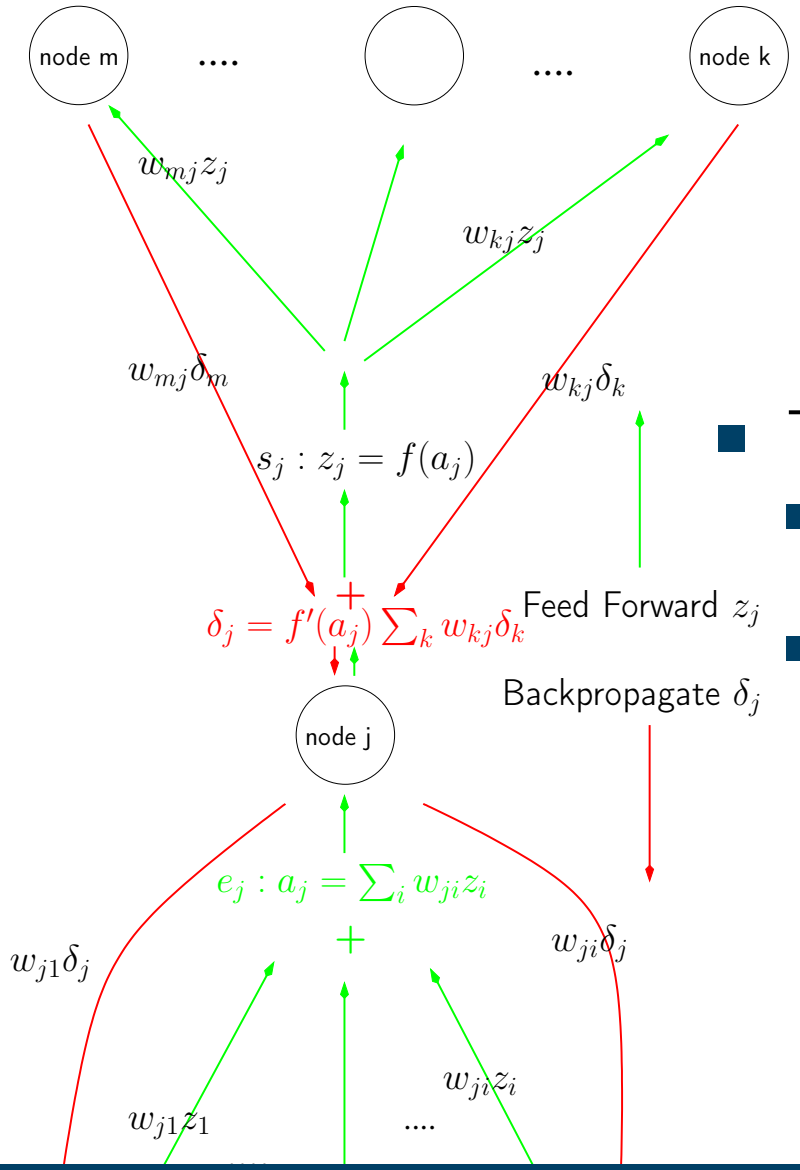
$$\frac{\partial E}{\partial a_j} = \sum_k \frac{\partial E}{\partial a_k}\frac{\partial a_k}{\partial a_j} = \sum_k \delta_k \frac{\partial a_k}{\partial a_j} \tag{4}$$

- Since: $a_k = \sum_i w_{ki} z_i = \sum_i w_{ki} f(a_i) \Rightarrow \frac{\partial a_k}{\partial a_j} = w_{kj} f'(a_j)$
- giving eventually that:

$$\delta_j = \frac{\partial E}{\partial a_j} = f'(a_j) \sum_k \delta_k w_{kj} \tag{5}$$

or in words: the error δ_j of a node j is the weighted sum of the errors of the nodes to which it passes its output.

Demonstration of MLP info-flow around the j node



- Training in MLPs goes in two steps:
  - Forward pass: inputs are passed forward in the net computing the a_ks and the z_ks
  - Backward pass: errors, δ_is, are passed backwards allowing for the computation of the gradient for every weight.

Backpropagation (on-line)

Initialise the weights
repeat
  for all $\mathbf{x}_n$ **do**
    Forward pass $\mathbf{x}_n$ and calculate the activations of the hidden units and the outputs.
    {Compute the δ_k of the output units (eq 3)}
    for all k output units **do**
      compute δ_k; $\Delta w_{kj} \leftarrow -\eta \delta_k z_j$
    end for
    {Back-propagate to compute the δ_j of the hidden units (eq 5)}
    for all j hidden units **do**
      Compute δ_j; $\Delta w_{ji} \leftarrow -\eta \delta_j z_i$
    end for
    {Update the weights }
    for all w_{ji} **do**
      $w_{ji} \leftarrow w_{ji} + \Delta w_{ji}$
    end for
  end for{for all $\mathbf{x}_n$}
until convergence

Backpropagation (batch)

Initialise the weights
repeat
  $\Delta w_{kj} \leftarrow 0$, $\Delta w_{ji} \leftarrow 0$
  for all $\mathbf{x}_n$ **do**
    Forward pass $\mathbf{x}_n$ and calculate the activations of the hidden units and the outputs.
    {Compute the δ_k of the output units (eq 3)}
    for all k output units **do**
      compute δ_k; $\Delta w_{kj} \leftarrow \Delta w_{kj} - \eta \delta_k z_j$
    end for
    {Back-propagate to compute the δ_j of the hidden units (eq 5)}
    for all j hidden units **do**
      Compute δ_j; $\Delta w_{ji} \leftarrow \Delta w_{ji} - \eta \delta_j z_i$
    end for
  end for{for all $\mathbf{x}_n$}
  {Update the weights}
  for all w_{ji} **do**
    $w_{ji} \leftarrow w_{ji} + \Delta w_{ji}$
  end for
until convergence

Online and batch backpropagation

- online:
  - We compute errors/derivatives and we adjust weights after seeing each example
  - Sensitive to the order of presentation of examples
  - Converges faster in the beggining but slower in general
  - The method of choise for very large datasets (stochastic gradient descent)
- Batch
  - We accumulate the derivative changes over the examples
  - We ajust the weights at the end of each cycle (epoch).

Plan

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2. Learning: backpropagation
3. **Optimisation and learning**

Initialisation of weights

- Objective : converge fast
- Initialise weights to random values
  - Uniformly distributed around 0 for the hiddent unitis
    $\rightarrow$ avoid extreme values on the sigmoid outputs
  - larger [-1,1] for the input units
    $\rightarrow$ if the weights w_{kj} are very smalle then δ_j and updates very small in the beggining
  - Initial weights should be different from one node to the other
    $\rightarrow$ if not all hidden units will do the same job

Learning rate η

Same discussion as in any gradient descent

- Value of η to be chosen experimentaly
- Very small: slow convergence
- Very large : oscilations, problematic convergences
- Rule of thumb : start with η large, and gradually reduce

Complexity of MLP backpropagation

- Computational complexity is

$$O(W)$$

where W = the number of parameters (weights and biases)

- Model complexity of a two layer MLP:

$$W = (d+1)h + (h+1)c$$

Model complexity control, MLP I

Reducing the number of hidden units

- Explore different numbers of hidden units and layers and choose the configuration that minimises the validation error
- Start with small numbers and increase during learning
- Start with a large network and prune
- Train a large number of small networks and combine

Model complexity control, MLP II

Regularisation : add a model complexity penalty

$$\tilde{E} = E + \Omega$$

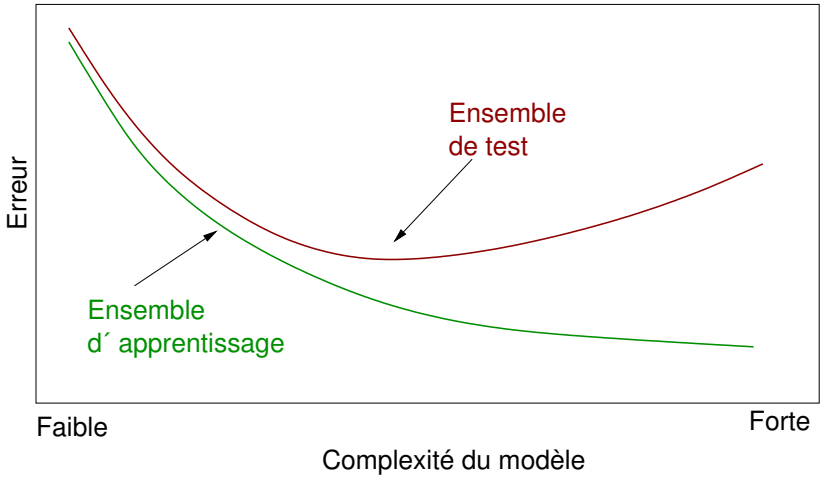
Ex. Weight decay

$$\Omega = \frac{1}{2} \sum_i w_i^2$$

To minimize $\tilde{E}$, we need to find the right trade-off between error (E) and model complexity Ω.

Model complexity control, MLP III

Early stopping : do not wait for convergence



- Fix the maximum number of epochs
- Stop if the training error does not decrease after k successive epochs
- Use a validation set, if validation error starts increasing stop.

MLP: summary

- can be used for classification and regression
- universal approximators
- robust to noisy
- we can generate arbitrary complex models
- training can be long
- regression and classification is fast
- have fallen to oblivion from the beggining of 90's
- back in fashion since 5 to 10 years now, known as deep neural networks (everybody is on them)
- big data, computational power and a number of tricks are responsible for the renaissance.

Books

- C. Bishop (2006). *Pattern Classification and Machine Learning*, Chapter 5: Neural Networks.
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